MUHAMMAD RIYADH AZHAR

Network Automation & Systems Engineer | Python, Go & Infrastructure Telemetry

Tangerang, Indonesia | +6289638855321 | mriyadhazhar@gmail.com | linkedin.com/in/mriazh/ | github.com/mriazh

Network Automation & Systems Engineer with 2+ years of experience bridging enterprise IT infrastructure, high-frequency network telemetry, and automated software pipelines. Proven track record building production-grade tools in Python and contributing security/routing enhancements to open-source Go infrastructure (LLM proxy routers). Experienced in developing asynchronous browser automation, SSH multi-vendor switch auditing, RouterOS API telemetry daemons, and OCR data pipelines backed by rigorous automated test suites (pytest, mypy, Go testing).

WORK EXPERIENCE

GMF Aeroasia (PT Garuda Maintenance Facility Aero Asia Tbk) | Tangerang, Indonesia
Network Support (April 2026 - Present)

- **Enterprise Fleet Auditing & Automation:** Architected automated multi-vendor switch audit engines (ArubaOS-CX, HP ProCurve) using Netmiko, collecting hardware health, PoE budgets, and VSF stack topologies across enterprise fleets to generate automated Jira Assets (CMDB) imports.
- **Operational Telemetry & Reporting Pipelines:** Engineered end-to-end automated reporting daemons in Python and Go, replacing manual SLA reporting with automated browser scraping, RouterOS API polling, and dual-engine OCR (PaddleOCR + Gemini fallback) with automated WhatsApp alerting.
- **Infrastructure Monitoring & Observability:** Maintained operational telemetry across 190+ hosts and 111+ enterprise switches using LibreNMS and PRTG, proactively diagnosing Layer 2/3 bottlenecks to ensure high availability for core enterprise services.
- **Access Control & IoT Security:** Enforced strict MAC-filtering policies for enterprise IoT perimeters (CCTV and environmental sensors) and managed segmented multi-SSID wireless environments across Huawei AirEngine Access Points.

Biznet (PT Supra Primatama Nusantara) | Bali, Indonesia
Network Operations Access Engineer (August 2024 - March 2026)

- **Diagnostics & Root-Cause Analysis:** Executed end-to-end optical anomaly diagnostics on enterprise GPON networks using OTDR and OPM tools, sustaining a 90% first-visit resolution rate on complex link impairments.
- **Standardization & Deployment:** Standardized configuration procedures for Customer Premises Equipment (CPE) across high-density FTTH deployments, reducing recurring misconfiguration tickets by 20%.
- **Technical Client Escalation:** Provided technical troubleshooting and onsite infrastructure resolutions, consistently achieving a 4.5/5 Customer Satisfaction (CSAT) rating.

Freelance / Self-Employed | Bali, Indonesia
Network Engineer (June 2025 - March 2026)

- **Large-Scale Switch Migration:** Spearheaded physical and logical migration of 28 Cisco Catalyst 9200L switches across 7 Bank BCA branches within strict maintenance windows, ensuring zero disruption to next-day financial operations.
- **Enterprise Mesh Optimization:** Designed and deployed multi-vendor mesh network topologies (Ruijie, TP-Link) for 10+ residential and commercial clients, expanding stable wireless coverage by 50%.

Zenius (PT Zona Edukasi Nusantara) | Jakarta, Indonesia
Full Stack Developer Intern (February 2022 - July 2022)

- **Full-Stack Web Development:** Developed web features using React.js and Node.js in an Agile sprint environment, contributing to user-facing learning portals, REST API integrations, and CI/CD code review processes.
- **Code Reliability & Delivery:** Participated in code reviews and cross-functional team sprints, ensuring clean component architecture and timely feature releases.

FEATURED PROJECTS

- **9router-go** | Go, Svelte, SQLite, Reverse Proxy
  Active core contributor to an open-source AI model proxy and high-concurrency LLM router, engineering SSRF security dialers, distributed tracing, database retention compaction, and multi-provider protocol translations.
- **Network-Daily-Report-Automation** | Python, Playwright, OpenPyXL, Systemd, GOWA Gateway
  Master orchestrator running 24/7 on Linux systemd, automating multi-portal telemetry scraping across 18 branch circuits, 12 modems, and enterprise APs with automated Excel compilation and daily WhatsApp broadcasts.

- **MRTG-CMP** | Python, RouterOS API, FastAPI, SQLite (WAL), Matplotlib, Leaflet
  Real-time network traffic telemetry system polling MikroTik branch routers over dedicated TCP tunnels, featuring RRDtool logarithmic autoscale graphing, multi-format reporting, and automated WhatsApp link alerts.
- **Switch-Collector** | Python, Netmiko, Pytest (328 tests), ArubaOS-CX/S, Jira CMDB, PyInstaller
  Multi-OS CLI automation auditing switch serial numbers, VSF stack chassis, PoE budgets, and fan health across enterprise switch fleets, featuring automatic syntax fallback and a 16-field Jira Assets CMDB export.

EDUCATION

UNIVERSITY OF JEMBER | Jember, Indonesia
Bachelor of Information Technology (Graduated 2024)
GPA: 3.47/4.00
Activities:
Kotlin Secretariat & IT Dev Public Relations at HIMATIF (2021), E-Sport Athlete (2019 - 2020).

CERTIFICATIONS

- **MikroTik Certified Network Associate (MTCNA)** - MikroTik (Mar 2026 - Mar 2029)
- **ACA Database Certification** - Alibaba Cloud (Nov 2024 - Nov 2026)
- **ACA Cloud Security Certification** - Alibaba Cloud (Oct 2024 - Oct 2026)
- **Exp: AWS Cloud Practitioner Essentials** - Dicoding Indonesia (Mar 2022 - Mar 2025)
- **Exp: Basic Azure Cloud** - Dicoding Indonesia (Feb 2020 - Feb 2023)

SKILLS

- **Programming & Automation:** Python, Go (Golang), Svelte, JavaScript, React.js, FastAPI, Node.js, Playwright, Selenium, Netmiko, Paramiko, REST APIs.
- **Testing & Code Quality:** Pytest (250+ unit/property tests), Go testing, Hypothesis, Mypy (Strict), Ruff, CI/CD (GitHub Actions).
- **Networking & Protocols:** TCP/IP, VLAN, STP, OSPF, LACP, RouterOS API, SSH Automation, GPON (OTDR/OPM), UTP/Fiber Cabling.
- **Infrastructure & Monitoring:** LibreNMS, PRTG, MRTG, Cisco Catalyst 9200L, ArubaOS-CX, HP ProCurve, Huawei WAC/AirEngine, MikroTik (MTCNA).
- **Databases & Systems:** SQLite (WAL Mode), Linux (Debian, Ubuntu), Windows, macOS.